ECON 6018 TUTORIAL

- By: Teresa Ng
- Lesson One

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Footnote: The practice questions are adapted from Essential Mathematics for Economic Analysis (5th Edition) by Knut Sydsaeter, Peter Hammond, Arne Strom, and Andres Carvajal. All credit for the original questions belongs to the authors. The solutions and explanations provided here are my own work.

Essential Materials

1. Textbook
   Essential Mathematics for Economic Analysis
   by Knut Sydsaeter, Peter Hammond, Arne Strøm
   & Andres Carvajal

- Fifth Edition

2. Writing Materials

- tutorials will be doing questions

3. Contact

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Questions:

1. Will we get these notes?
- Yes, they will be sent out 2 days after lecture

2. How does tutorial help?
- help you get familiar with concepts
- majority of students who attended all my tutorial last sem got B+ and above

Any others?

1. System of Linear Equations

In Lecture:

Eg.

$$2x_1 + 3x_2 + 4x_3 = 1$$
$$3x_1 + 4x_2 + 5x_3 = 2$$
$$4x_1 + 5x_2 + 6x_3 = 3$$

- can be represented as

$$\underbrace{\begin{bmatrix} 2 & 3 & 4 \\ 3 & 4 & 5 \\ 4 & 5 & 6 \end{bmatrix}}_{\substack{A \\ 3\times 3}} \times \underbrace{\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}}_{\substack{x \\ 3\times 1}} = \underbrace{\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}}_{\substack{d \\ 3\times 1}}$$

$$A \times x = d$$

$3\times3 \quad 3\times1 \quad 3\times1$

coefficient matrix

Practices:

Step 1: Write all the common terms in the same column

- there may be alot of variables so this makes sure we are not confused

Step 2: Seperate the coefficients

Practice Questions:

Chpt 15.1

Q1.

$$3x_1 - 2x_2 + 6x_3 = 5$$

$$5x_1 + x_2 + 2x_3 = -2$$

Q2.

(i) $C = 0.712y + 95.05$

(ii) $y = C + x - S$

(iii) $S = 0.158(C + x) - 34.30$

(iv) $x = 93.53$

Answer:

Q1. $$\begin{pmatrix} 3 & -2 & 6 \\ 5 & 1 & 2 \end{pmatrix}\begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 5 \\ -2 \end{pmatrix}$$

Q2. $$\begin{pmatrix} 0 & -0.712 & 1 & 0 \\ -1 & 1 & -1 & 1 \\ -0.158 & 0 & -0.158 & 1 \\ 1 & 0 & 0 & 0 \end{pmatrix}\begin{pmatrix} x \\ y \\ c \\ s \end{pmatrix} = \begin{pmatrix} 95.05 \\ 0 \\ -34.30 \\ 93.53 \end{pmatrix}$$

Extra

How is this use in Economics?

- PhD Level Macro Economics Modelling
  "Job Search Model"
- Value of being unemployed
  Value of being employed
  wage

$\left.\begin{matrix} \text{Value of being unemployed} \\ \text{Value of being employed} \\ \text{wage} \end{matrix}\right\}$ solving for

allow $x = \begin{bmatrix} V_u \\ V_e \\ w \end{bmatrix}$

$$V_u = (b-\kappa) + \beta[(1-p)V_u + pV_e]$$
$$V_E = w + \beta[(1-\delta)V_E + \delta V_u]$$
$$w = \phi y + (1-\phi)b$$

$$\begin{bmatrix} 1-\beta(1-p) & -\beta p & 0 \\ -\beta\delta & 1-\beta(1-\delta) & -1 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} V_u \\ V_e \\ w \end{bmatrix} = \begin{bmatrix} b-\kappa \\ 0 \\ \phi y + (1-\phi)b \end{bmatrix}$$

2. Matrix Operations

In lecture: (Addition and subtraction)

(a) $(A+B)+C = A+(B+C)$

(b) $A+B = B+A$

(c) $A+0 = A$

(d) $A+(-A) = 0$

(e) $(\alpha+\beta)A = \alpha A+\beta A$

(f) $\alpha(A+B) = \alpha A+\alpha B$

Practices

Chpt 15.2

Q1. $A = \begin{pmatrix} 0 & 1 \\ 2 & 3 \end{pmatrix}$ $B = \begin{pmatrix} 1 & -1 \\ 5 & 2 \end{pmatrix}$

- $A+B$
- $3A$

Q2. $A = \begin{pmatrix} 0 & 1 & -1 \\ 2 & 3 & 7 \end{pmatrix}$ $B = \begin{pmatrix} 1 & -1 & 5 \\ 0 & 1 & 9 \end{pmatrix}$

- $A+B$
- $A-B$
- $5A-3B$

Ans:

Q1. A+B $\begin{pmatrix} 1 & 0 \\ 7 & 5 \end{pmatrix}$

$3A = \begin{pmatrix} 0 & 3 \\ 6 & 9 \end{pmatrix}$

Q2. A+B $\begin{pmatrix} 1 & 0 & 4 \\ 2 & 4 & 16 \end{pmatrix}$

A-B $\begin{pmatrix} -1 & 2 & -6 \\ 2 & 2 & -2 \end{pmatrix}$

5A-3B $\begin{pmatrix} -3 & 8 & -20 \\ 10 & 12 & 8 \end{pmatrix}$

In Lecture: Matrix Multiplication

$A \times B = AB$

Iff

A is $m \times n$

B is $n \times m$

Final product: $m \times m$

Eg. $A = \begin{pmatrix} 0 \\ 2 \end{pmatrix}$ $B = \begin{pmatrix} 1 & 2 \end{pmatrix}$

2×1 1×2

$AB = \begin{pmatrix} 0 & 0 \\ 2 & 4 \end{pmatrix}$

Practices

Chpt 15.3

Q1. Calculate AB and BA

(a) $\begin{pmatrix} 0 & -2 \\ 3 & 1 \end{pmatrix}$ and $\begin{pmatrix} -1 & 4 \\ 1 & 5 \end{pmatrix}$

(b) $\begin{pmatrix} 8 & 3 & -2 \\ 1 & 0 & 4 \end{pmatrix}$ and $\begin{pmatrix} 2 & -2 \\ 4 & 3 \\ 1 & -5 \end{pmatrix}$

Answer

Q1. (a)

$AB \quad \begin{pmatrix} -2 & -10 \\ -2 & 17 \end{pmatrix}$

$BA \quad \begin{pmatrix} 12 & 6 \\ 15 & 3 \end{pmatrix}$

(b)

$AB \quad \begin{pmatrix} 26 & 3 \\ 6 & -22 \end{pmatrix}$

$BA \quad \begin{pmatrix} 14 & 6 & -12 \\ 35 & 12 & 4 \\ 3 & 3 & -22 \end{pmatrix}$

In Lecture: Rules for Matrix Multiplication

$$(AB)C = A(BC)$$
$$A(B+C) = AB + AC$$
$$(A+B)C = AC + BC$$
$$(\alpha A)B = A(\alpha B) = \alpha(AB)$$

Power matrix — square matrix

$$A^2 = A \times A$$
$$A^n = \underbrace{AA \ldots A}_{n \text{ times}}$$

Practices:

Chpt 15.4

Suppose P, Q ae $n \times n$ matrices

$PQ = Q^2 P$

Prove that $(PQ)^2 = Q^6 P^2$

Ans:

$$PQ = PQPQ$$
$$= Q^2 PPQ$$
$$= Q^2 P(Q^2 P)$$
$$= Q^2 PQ\ QP$$
$$= Q^2 (Q^2 P) QP$$
$$= Q^4 (PQ) P$$
$$= Q^4 (Q^2 P) P$$
$$= Q^6 P^2$$

In Lecture:

Identity Matrix

$$\mathbf{I}_n = \begin{pmatrix} 1 & 0 & \dots & 0 \\ 0 & 1 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & 1 \end{pmatrix}_{n \times n}$$

$$\mathbf{A}\mathbf{I}_n = \mathbf{I}_n\mathbf{A} = \mathbf{A}$$

Practices:

Chpt 15.4

Qns 1.

Prove (i) $(A+B)(A-B) \neq A^2 - B^2$

(ii) $(A-B)(A-B) \neq A^2 - 2AB + B^2$

A square matrix $\mathbf{A}$ is said to be *idempotent* if $\mathbf{A}^2 = \mathbf{A}$.

(a) Show that the matrix $\begin{pmatrix} 2 & -2 & -4 \\ -1 & 3 & 4 \\ 1 & -2 & -3 \end{pmatrix}$ is idempotent.

(b) Show that if $\mathbf{AB} = \mathbf{A}$ and $\mathbf{BA} = \mathbf{B}$, then $\mathbf{A}$ and $\mathbf{B}$ are both idempotent.

(c) Show that if $\mathbf{A}$ is idempotent, then $\mathbf{A}^n = \mathbf{A}$ for all positive integers n.

Qns 2.

Idempotent if $A^2 = A$

(a) Show $\begin{pmatrix} 2 & -2 & -4 \\ -1 & 3 & 4 \\ 1 & -2 & -3 \end{pmatrix}$ is idempotent

(b) If $AB = A$, $BA = B$, show A and B are idempotent

(c) Show $A^n = A$ for all positive integers n

Qns 3. If $P^3 Q = PQ$. Prove $P^5 Q = PQ$

Answers

Qns 1:

(ai) $(A+B)(A-B) = A(A-B) + B(A-B)$

$= A^2 - AB + BA - B^2$

$\neq A^2 - B^2$

(aii) $(A-B)(A-B) = A^2 - AB - BA + B^2$

$\neq A^2 - 2AB + B^2$

Q2. (a) $A^2 = A$

(b) $AB = A = AA$

$B = A$

$BA = B$

$BB = B$

$B^2 = B$ (Shown)

(c) $A^n = A$ when $n = 1$

Induction

$A^k = A$

$A^{k+1} = A^k A = AA = A$

Q3. $P^5 Q = P^2 P^3 Q$

$= P^2 (PQ)$

$= P^3 Q$

$= P$

3. Transpose

In Lecture:

$$(A')' = A$$

$$(A+B)' = A' + B'$$

$$(\alpha A)' = \alpha A'$$

$$(AB)' = B'A'$$

Practice Question

Chpt 15.5

Q1. Find transpose

Find the transposes of $\mathbf{A} = \begin{pmatrix} 3 & 5 & 8 & 3 \\ -1 & 2 & 6 & 2 \end{pmatrix}$, $\mathbf{B} = \begin{pmatrix} 0 \\ 1 \\ -1 \\ 2 \end{pmatrix}$, and $\mathbf{C} = \begin{pmatrix} 1 & 5 & 0 & -1 \end{pmatrix}$.

$$A = \begin{pmatrix} 3 & 5 & 8 & 3 \\ -1 & 2 & 6 & 2 \end{pmatrix} \qquad B = \begin{pmatrix} 0 \\ 1 \\ -1 \\ 2 \end{pmatrix} \qquad C = \begin{pmatrix} 1 & 5 & 0 & -1 \end{pmatrix}$$

Q2. Show that

$$(A_1 A_2 A_3)' = A_3' A_2' A_1'$$

Ans:

Q1. $A' = \begin{pmatrix} 3 & 1 \\ 5 & 2 \\ 8 & 6 \\ 3 & 2 \end{pmatrix}$

$B' = \begin{pmatrix} 0 & 1 & 1 & 2 \end{pmatrix}$

$C' = \begin{pmatrix} 1 \\ 5 \\ 0 \\ -1 \end{pmatrix}$

Q2. $(A_1 A_2 A_3)' = (A_3)'(A_1 A_2)'$

$= A_3' A_2' A_1'$

$=$